Towards INFO-RAG on the edge

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1 Introduction

Retrieval-Augmented Generation (RAG) is a framework widely used by today's Large Language Models (LLMs) to provide high-quality responses to user queries. Customer support assistants, AI chatbots, and search engines are implemented through RAG. The traditional technique can be thought of in two steps: retrieval and generation. In retrieval, numerous databases are scanned to extract valuable and relevant information. In generation, the retrieved information is taken as context and then integrated with the user query to generate a reliable response.

The current issue with RAG is that traditional LLM architecture may be passively collecting and combining information rather than actively understanding it. We can see this in casual use of AI chatbots—these models often misinterpret contradictory information, ignore important details, and fail to combine multiple sources.

Xu et al. 2024 sought to mitigate this problem by implementing an unsupervised fine tuning method, InfoRAG, which introduces an objective called "positive information gain" for LLMs to consider when extracting information. This objective measures how useful and relevant a source's information is to the query. InfoRAG was trained on an unsupervised dataset of unlabeled Wikipedia articles.

So far, InfoRAG has been successfully evaluated on LlaMa 2 with 7-10B parameters. However, a question arises as to whether this framework can be applied to smaller models. Our project's objective is to fine tune InfoRAG on smaller models capable of being stored in edge devices such as smartphones. Parameter sizes for smartphone LLMs can vary, but are generally 7B or less. Apple's on-device LLM tools for iPhone and iPad use 3B parameters (Apple Machine Learning Research, 2024).

2 Methods

2.1 Preliminary

Low-Rank Adaptation (LoRA) is a parameter efficient fine-tuning method that works by freezing the weights of a pre-trained model and inserting trainable matrices into a model's attention layers (Hu et al., 2021). Rather than just directly updating all of a model's parameters, LoRA decomposes weight updates into smaller matrices which significantly cuts down on the number of parameters that need to be trained. This helps reduce the compute need for fine tuning, making the project meaningfully more computationally feasible given our resources.

2.2 Data Construction

Replicating the procedure outlined by Xu et al. 2024 in the original paper, we construct a form of unsupervised training data out of the DPR Wikipedia corpus (Karpukhin et al., 2020). For each Wikipedia article we extract sets of k-consecutive sentences. From each set of sentences a single sentence is randomly selected to be the target sentence. This target sentence is then split into two halves, a prefix and a suffix. This prefix-suffix set up aims to simulate a query and answer, and the other non-target sentences in the set act as the retrieved documents, where the answer is embedded.

After completing this construction, we generate training data in the structure of three scenarios.

The first scenario has retrieved documents that have all the necessary information for answering the query, and the model is trained to locate and extract the correct target information. In the second scenario, named entities within the retrieved documents are randomly masked as a method to simulate incomplete or misleading information. This requires the model to learn to rely on surrounding information to fill in the gaps. The third scenario has the target scenario completely removed from the retrieved documents, which forces the model to use context clues and previously learned knowledge to answer the query.

As in the original paper, we dedicate 20% of the training data to the first scenario, and 40% to the second and third scenarios each. In total, we construct 10,000 training samples.

2.3 Model Selection & Finetuning

Using our replicated InfoRAG fine tuning paradigm, we trained a total of 9 LLMs of ranging architectures and parameter sizes:

- Families: LLaMa, Qwen, Gemma, Phi, SmolLM
- Parameter Sizes: 0.5B, 1B, 1.5B, 2B, 3B

Selecting these models allowed us to test the information refinement technique on a wider range of families, whereas the original paper only looked at the LlaMa family. Furthermore, our wide range of parameter sizes ensured that results would replicate even to smaller models, with potential to explore whether parameter size modulated finetuning improvements.

Crucially, we maintained the same hyperparameters for all models. This choice allows us to claim that any boost in benchmarks was a result of the finetuning method itself, not a difference in hyperparameters. We acknowledge that deployment of models trained under the InfoRAG technique should be hyperparameter tuned to ensure effectiveness in downstream tasks. However, this project was solely to replicate results, not create the best performing model. Avoiding any confounding variables in our results, the following hyperparameters were used for all models:

- Method: LoRA with learning rate of 1e-5
    - r = 16
    - alpha = 16
- Batch Size: 2 per device * 8 gradient accumulation = 16
- Training Steps: 5,000
    - Evaluated every 1,000 steps

3 Experiments

3.1 Experimental Setup

Given that the original paper evaluates on over ten benchmark datasets, many of which are not easily accessible or take multiple steps to initialize, we evaluate our models on only a subset of benchmarks from the original paper. The first of these selected benchmarks is HotpotQA (Yang et al., 2018). HotpotQA is a multi-hop question answering benchmarking dataset, which requires reasoning over multiple documents to extract the response to a question. The second of these benchmarks is Wizards of Wikipedia (Dinan et al., 2019). Wizards of Wikipedia is a dialogue-based benchmarking dataset that evaluates a model's ability to extract information from information stored in chat format, as might be typically encountered by a chatbot. Use of these two benchmarking datasets eliminated the need to set up a full separate RAG pipeline during evaluation, making the scope of the project more feasible given our time constraints.

Following Xu et al. 2024 in their evaluation setup, we use Exact Match (EM) as the primary metric for HotpotQA, and token-level F1 for evaluating results on Wizards of Wikipedia. Exact Match works by checking if the model's output matches the expected output exactly, while token-level F1 measures the overlap in tokens between the model's output and the expected response. Each metric is chosen to best complement the nature and structuring of the accompanying benchmark.

We evaluate each model with each benchmark at multiple training step checkpoints, scoring with both metrics. Performance is then compared against the baseline non-fine-tuned models to determine the effectiveness of fine-tuning.

3.2 Results & Analysis

Our replicated results show that the InfoRAG training technique improves models performance even given our subset of RAG benchmarks.

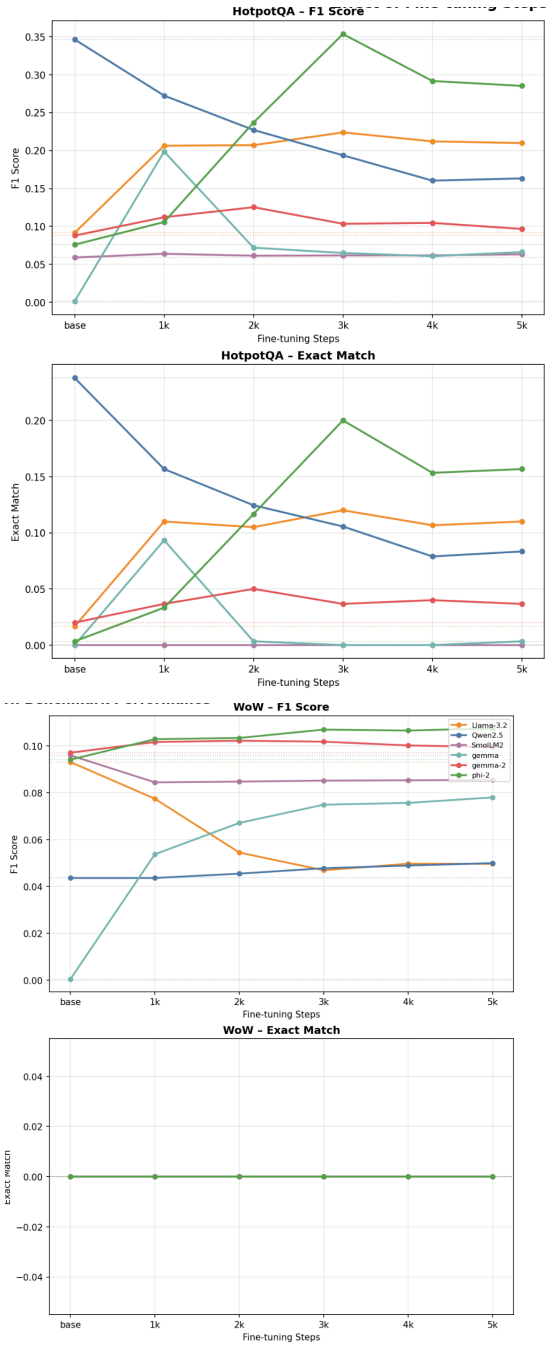
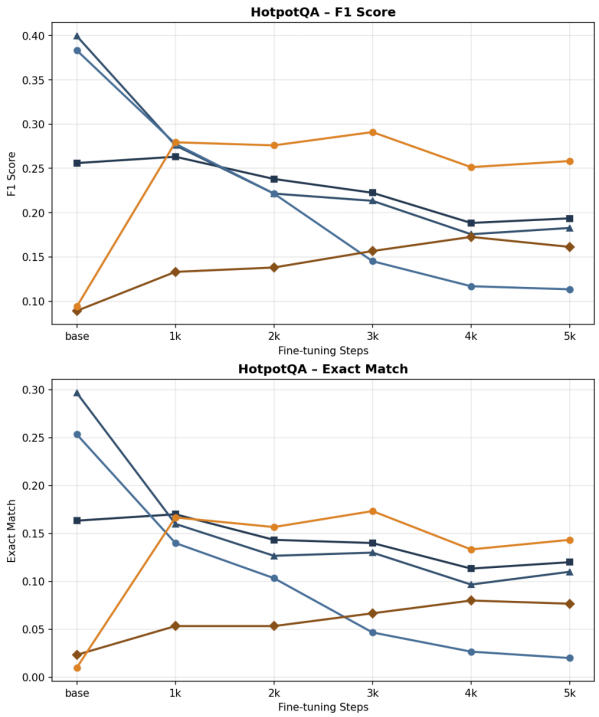


Fig 1a-1d. Effect of Fine-Tuning Steps on Benchmark Performance

Figure 1a-1d shows model performance on each benchmark metric with respect to the amount of finetuning steps. Evaluation at zero steps is effectively the base model evaluated on the benchmark. Generally, models show an upward trend in at least one of the metrics. That is, for each model, there is at least one metric where performance increased some steps after fine tuning compared to the baseline performance. Although the exact match wasn't used in the WoW benchmark, we included it for the sake of completeness -- although the flatline results don't tell us anything informative. The takeaway here is that while different models handle the fine tuning differently in terms of its impact on the evaluation performance, InfoRAG finetuning increases performance on RAG benchmarks.



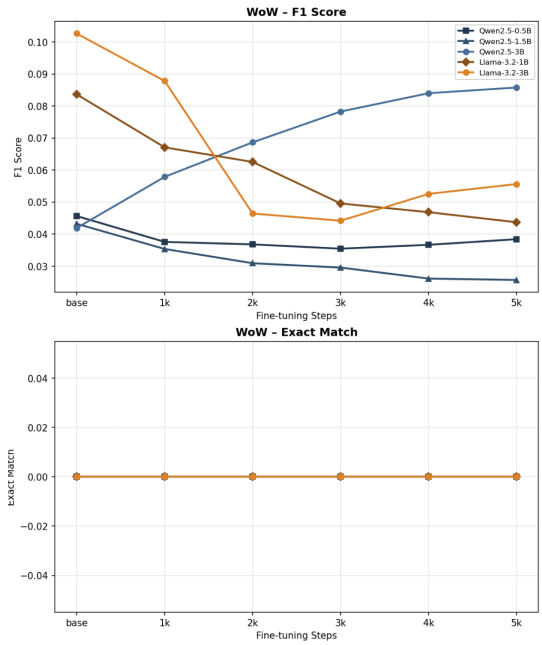


Fig 2a-2d. Effect of Parameter Size on Benchmark Performance

Figure 2a-2d highlights our LlaMa and Qwen families of models, specifically their different parameter sizes. For LlaMa, we tested the 1B and 3B parameter models. Across all relevant metrics, we see that the higher parameter model not only performs better on the RAG benchmarks, but also shows more improvement after InfoRAG fine tuning compared to the lower parameter model. This can be explained through the concept of catastrophic forgetting, where lower parameter models may be more prone to dramatic changes in model performance after fine tuning. The Qwen models show an interesting case where model performance overall decreased after finetuning. Although the Qwen 3B parameter model does steadily increase in performance after fine tuning on the WoW F1 benchmark, this family of model's performance on other benchmarks shows signs that model family may impact the effect of fine tuning. Conclusively, this figure shows that the LlaMa model handles InfoRAG finetuning better than the Qwen models do, although both families do show statistically significant increase in performance after finetuning -- more on statistical significance in Figure 4.

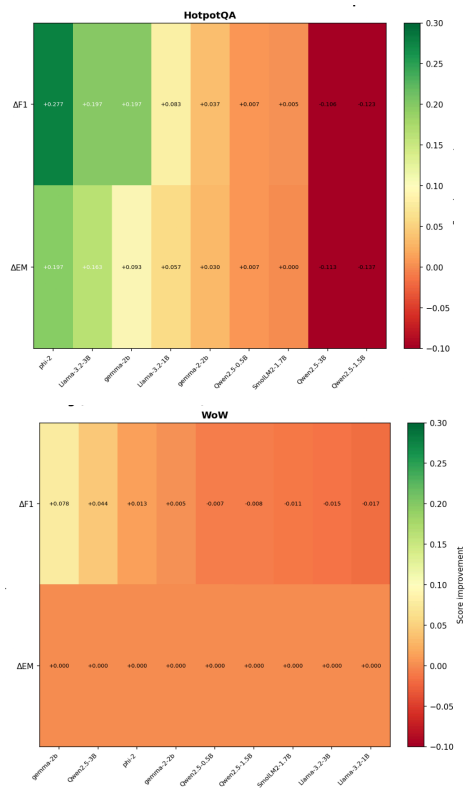


Fig 3a, 3b. Maximum improvement from fine-tuning (Base → Best Tuned)

To explore which models improved the most after finetuning, we ranked the difference in performance (delta F1, delta EM) across all models in Figure 3a-3b. In general, the HotPotQA benchmark increased the most as a result of this fine tuning, an interesting result considering our fine tuning data contained Wikipedia articles which are also used in the WoW benchmark during evaluation. Nevertheless, the increased performance on the HotPotQA benchmark indicates that the increase in performance is indeed due to the information refinement training objective, not due to data leakage or parameter activation due to fine tuning on a contextually similar dataset. Surprisingly, the most gain in performance after training was seen in the Phi-2, Llama, and Gemma models, a sign that we've successfully replicated the InfoRAG improvements within Llama and extended these findings into models outside of the family of models the original paper used.

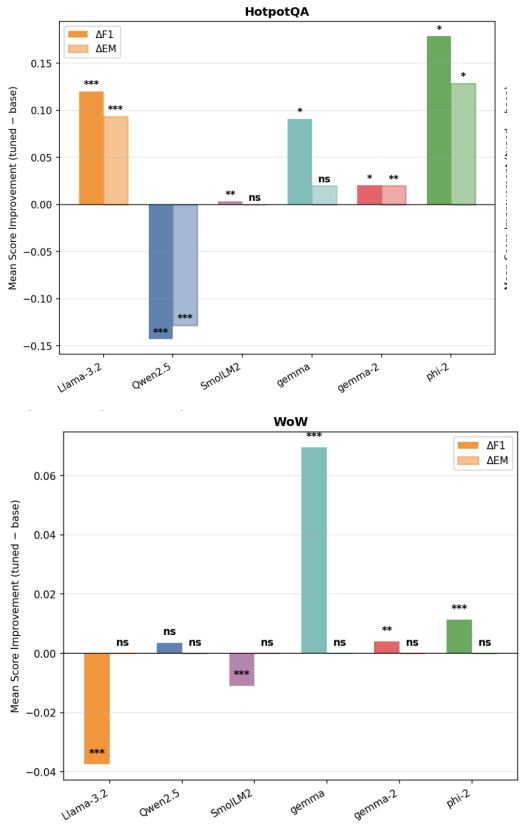


Fig 4a, 4b. Statistical significance of fine-tuning improvement (one sample t-test vs. baseline; *p<.05, **p<.01, ***p<0.001)

For statistical concreteness, we then conducted a one-sample t-test across all models of the baseline performance compared to the improved performance, shown in Figure 4a-4b. Our results show that on the HotPotQA benchmark, LlaMa, Gemma, Gemma2, and Phi-2 had statistically significant improvements in token level F1 ($p<0.05$). On the WoW benchmark, only Phi-2 and Gemma increased in performance for token level F1 ($p<0.001$). We conclude that the models that improved the most were concretely the Phi-2 and Gemma models. Yet, models showed improvements in specified areas. Thus, we confidently say that most models improved due to InfoRAG training, with some noise in the performance results canceling out if we would've had the bandwidth to evaluate the models on more benchmarks.

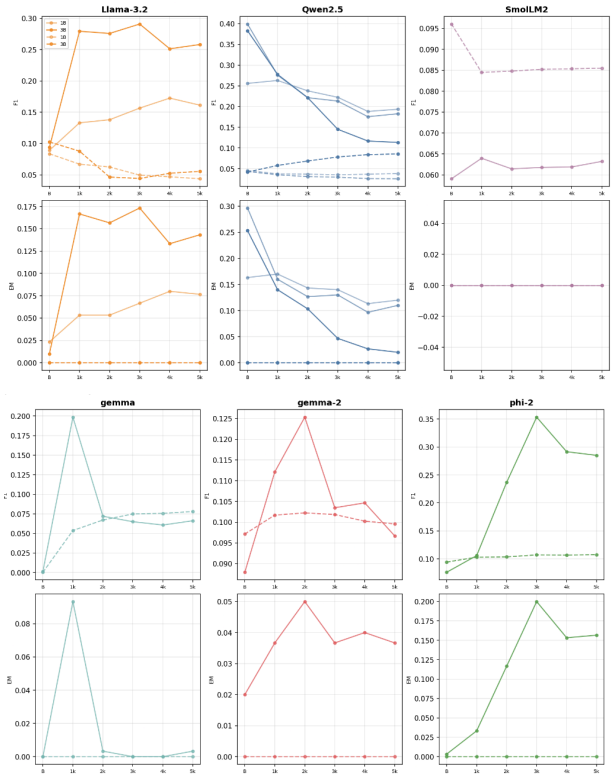


Fig 5a-5f. Step-by-Step F1 Progression per Model Family (solid = HotpotQA, dashed = WoW)

Our final figure, Figure 5a-5f, shows performance on each benchmark as a function of training steps. Each model had a different training step where it achieved the highest improved performance after finetuning. To reiterate, we believe that deploying these models would require proper hyperparameter tuning over these benchmarks to figure out the best parameters to train these models on (learning rates, batch sizes, gradient accumulation steps, training steps, etc...). Our focus wasn't to create the best model, but to show that RAG performance increased after finetuning. Indeed, most models showed an increase in performance after finetuning, but many required less training steps for peak performance than the training steps performed by the original paper (the full 5,000).

4 Conclusion & Future Directions

Our replication shows that the InfoRAG training performance indeed extends to not only smaller parameter sizes of models, but also different families of models the original paper didn't use. While our results show some noise in RAG

performance after fine tuning, we believe there is significant evidence that improvement happened as a result of the unsupervised training method. At face value, our project shows potential for the application of this training objective on smaller parameter models, models that may be used in edge or limited compute situations.

Many questions remain despite our replication. While we tested smaller parameter models compared to the original paper, the opposite direction could be done to validate if increased RAG performance occurs in higher parameter models as well. Furthermore, more benchmarks could be used to validate our replicated results, perhaps seeing if increased improvements occur even outside of traditional RAG benchmarks. Finally, InfoRAG only considers changes to the LLMs for increased RAG performance, but more work needs to be done to evaluate analogous versions of this strategy on retrieval systems, or even LLMs with built in retrieval. Therefore, our successful replication highlights a new frontier for further experimentation to be done using this data augmentation technique.

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